Bradley Schalk

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US Citizen | Eligible for US Security Clearance | Willing to Relocate

EDUCATION

University of Illinois Urbana-Champaign **May 2029**
B.S. Aerospace Engineering, Minor in Business

SKILLS & TECHNICAL TOOLS

Manufacturing: Fusion CAM, CNC machining, milling, workholding, design for manufacturing (DFM), 3D printing
CAD & Design: Siemens NX, Fusion 360, mechanical design, assemblies, reverse engineering
Robotics & Automation: FANUC RoboGuide, teach pendant (TP) programming, industrial automation
Controls & Programming: PLC ladder logic, I/O, sensors, Python, Java, Microsoft Excel, Microsoft 365
Other: Mechanical troubleshooting, forklift certified, root cause analysis, technical documentation

EXPERIENCE

Perasmic — Business Development & Customer Discovery | Urbana-Champaign, IL **Oct 2026 – Present**

- Conduct outreach to independent retailers using KORONA POS, communicating directly with prospective users to identify operational pain points, gather pre-launch feedback, and evaluate how they would use an AI-powered POS analytics service.
- Support early-stage marketing and go-to-market efforts by gathering and relaying customer feedback to inform product and messaging priorities ahead of the company's paid launch.

Manufacturing Engineering Intern — Metamora Industries | Metamora, IL **May 2026 – Aug 2026**

- Modeled a custom EOAT plate in Fusion 360, generated Fusion CAM toolpaths, CNC-machined the plate, and assembled it with a pneumatic piston, sensor, and magnetic gripper; iterated geometry to eliminate robot collision risk.
- Reverse-engineered heat-treat furnace conveyor components in Fusion 360; generated Fusion CAM toolpaths and custom workholding for replacement rollers, links, pins, bosses, and sprockets used in a full conveyor rebuild.
- Hand-measured a fractured furnace component, rebuilt it to exact dimensions in Fusion 360, and produced a manufacturing-ready CAD model used as the dimensional basis for CNC machining its replacement component.
- Programmed and commissioned a FANUC R-1000iA/100F robot to move track shoes among a conveyor, Yama Seiki BM-1600MAX VMC, and finished-part racks; taught hundreds of TP positions and wrote four product-specific routines using RoboGuide and teach-pendant programming, with the automated cell projected to improve production efficiency by approximately 30%.
- Wrote PLC ladder logic and configured I/O among the FANUC robot, VMC, light curtains, door sensors, and part-presence sensors to sequence autonomous cycles and enforce safety interlocks.
- Traced control-cabinet wiring to troubleshoot and diagnose machine states throughout automated cell operation.

PROJECTS

Custom FPV Drone — Personal Project **Aug 2026 – Present**

- Building a 3-inch FPV quadcopter from discrete components, selecting and integrating brushless propulsion, flight controller/ESC, ELRS radio receiver, analog video transmission, and LiPo power systems into a complete airframe.

B-36 Peacemaker — Siemens NX Aircraft Model **Nov 2025 – Dec 2025**

- Built a high-fidelity Convair B-36 Peacemaker in Siemens NX, modeling the fuselage, wings, propulsion systems, landing gear, and major aircraft components within a fully constrained digital assembly.
- Applied assembly constraints to moving mechanical components and created a rendered engineering animation demonstrating mechanical motion throughout the completed aircraft assembly.

Mechanically Launched Glider — Team Lead **Oct 2025 – Nov 2025**

- Led a three-prototype design-build-test cycle for a mechanically launched glider, modifying airframe geometry and trim between builds to improve stability; achieved a 179 ft flight distance and 6.35 s flight duration.

Model Rocket Design & Flight Test **Sep 2025 – Oct 2025**

- Configured and iteratively refined a model rocket in OpenRocket to target a 325 ft apogee, then built and flight-tested the completed vehicle to approximately 360 ft while comparing simulated and observed flight performance.

LEADERSHIP & INVOLVEMENT

Triangle Fraternity — Merchandise Coordinator, Rush Chair, Social Chair **Aug 2025 – Present**

- Led event logistics, vendor coordination, merchandise procurement, and recruitment efforts as Merchandise Coordinator, Rush Chair, and Social Chair, supporting chapter activities and executive leadership responsibilities.